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Education

July 2022 — **PhD, University of California, San Diego** in Physics
Thesis Title: *The Properties and Evolution of Star Forming Regions Over Cosmic Time*
Thesis Advisor: Prof. Shelley A. Wright

January 2020 — **C.Phil., University of California, San Diego** in Physics
Thesis Advisor: Prof. Shelley A. Wright

June 2016 — **B.S., California Polytechnic State University, San Luis Obispo** in Mechanical Engineering
Minors: Physics, Astronomy — Honors: *Cum Laude*

Research Experience

2022-present — **Brinson Prize Postdoctoral Fellow in Astronomical Instrumentation**, Carnegie Observatories, Pasadena

Magellan Infrared Multi-Object Spectrograph (MIRMOS)
Instrument Scientist & Subsystem Lead

- Coordinating between the science and various technical teams to ensure the design of this complex vacuum cryogenic instrument provides the necessary performance to meet or exceed the scientific requirements
- Overseeing development of Assembly, Integration, and Testing (AIT) plan for the full instrument and subsystems
- Leading development of slicer style integral field unit (IFU) and atmospheric dispersion corrector (ADC): performing optical design and analysis; coordinating vendor selection and contracts; coordinating mechanical engineering effort and system requirements; developing requirements for necessary data reduction and analysis software
- Overseeing documentation repository to ensure version history is maintained and required design documentation is generated

Local Volume Mapper (SDSS-V)
Working Group Chair

- Leading work on key collaboration science priorities including a catalog of all ionized nebulae in the Magellanic Clouds, a study of stellar feedback in these catalog regions, and a pilot study of resolved stellar feedback in the SMC's largest star forming region
- Led bi-weekly telecons to discuss and coordinate ongoing science projects
- Organized early science targets for commissioning
- Spectrograph lead for proposed follow-up instrument and survey at Mcdonald Observatory 2.1m telescope

Magellan Survey of Ionized Gas Kinematics in Nearby Galaxies

- Leading a survey of local group dwarf galaxies using the new optical integral field spectrograph at Magellan, IFU-M
- Designed target selection to include ∼10 galaxies over a wide range of stellar mass, star formation rate, and metallicity to uncover the impact of environment on stellar feedback

Research Experience (continued)

2016-2022 **Graduate Student Researcher**, Physics Department / Center for Astrophysics and Space Sciences (CASS), University of California, San Diego
Advisor: Prof. Shelley A. Wright

- **Observations of Star-forming Regions**
  - Studied the evolution of galaxies and star formation through Keck Observatory Integral Field Spectrograph observations (KCWI and OSIRIS) of the gaseous regions which have undergone recent star formation
  - Led observing proposals, plannning and data reduction/analysis using a combination of public pipelines and custom routines (primarily Python based, secondarily IDL)
  - Studied the ionization states and kinematics of these regions, the impact of star formation feedback, and the evolution of star forming regions over time
- ***Liger* Imager and IFS**
  - Completed design of the filter wheels, pupil wheel, and detector focus stage with associated stress and deflection analysis, and obtained preliminary quotes
  - Coordinated with team members in other specialties (e.g, optical design, electrical) to ensure requirements are met and interfaces will be functional
  - Contributed to major funding proposals including obtaining quotes, generating photo-realistic part models, and text
- **Panoramic SETI**
  - Assisted with development and testing of prototype telescope module design for an all-sky SETI observatory. Designed lens mounting and module baffling
  - Worked on characterization of near-infrared discrete amplification photon detector and integration with existing readout electronics

2014 – 2016 **Student Researcher**, Physics Department, California Polytechnic State University, San Luis Obispo
Advisor: Prof. Vardha N. Bennert

- **Local Active Galactic Nuclei (AGN)**
  - Used emission-line spectra of active galaxies to gain insight into the AGN phenomenon (particularly the FeII emissions, Hβ emission line variability, and mass scaling relations)
  - Assisted new students in fitting spectra and programming in IDL by creating a detailed guide to fitting a range of AGN spectra as well as direct mentoring of new students

Teaching & Mentoring

Courses

Winter 2019 **TA: Galaxies and Quasars (Physics 163)** at UC San Diego

- Led discussion sections with additional lecture material and example problems
- Developed additional example problem sets for exam preparation and a guide to writing a term paper

Fall 2016 **TA: Electricity and Magnetism Laboratory (Physics 1BL)** at UC San Diego

- Led three lab sections, including administering quizzes, conducting brief lectures, and guiding experiments

Fall 2013 **TA: General Chemistry for Physical Science and Engineering** at Cal Poly, SLO

- Assisted with laboratory experiments and grading lab reports

Teaching & Mentoring (continued)

Workshops

Summers 2023-26 **Carnegie Astrophysics Summer Student Internship (CASSI)**

- Taught workshops on programming with Python, data visualization, scientific writing, responsible use of AI, and applying to graduate school for student interns
- Developed a new, hands-on instrumentation workshop to add to the programs' complement of educational activities

2021 **AstroTech Instructor**

- Co-developed and led inquiry based activity teaching basic optics as introduction to instrumentation workshop for early career scientists
- Facilitated additional optics activity teaching image formation and operation of lab equipment
- Participated in career panel for summer school participants

2018 **Professional Development Program**

- Worked with a team of three scientists to develop an inquiry based activity for summer undergraduate researchers at UCSD using telescope simulators to teach research skills and key astronomical concepts (e.g., angular resolution)
- As preparation, participated in an intensive 5-day workshop on inquiry based learning, inclusive teamwork, and facilitation techniques

Mentoring

Summers 2023,26 **Carnegie Astrophysics Summer Student Internship (CASSI)**

- 2026: Worked with a summer intern from a nearby community college to determine the strength of stellar feedback in Magellanic Cloud H II regions from LVM
  - As their first research experience, this involved guiding them through the scientific background, programming and data analysis techniques, and scientific communication
- 2023: Worked with a summer intern from a Cal State University (CSU) on a project to determine the mechanical precision required to position key optical components of MIRMOS
  - Required guiding them through the basics of optics, tolerance analysis in Zemax, fitting data in python, and communicating results in written and oral formats

July 2024-25 **Astronomy Mentoring Program for Upcoming Postdocs (AMP-UP)**

- Mentored 2 graduate students to help them navigate the job market with an emphasis on the postdoctoral fellowship application process
- The program aims to increase the availability of peer to peer mentoring beyond the institutions which already host a significant number of fellows

May/June 2023 **Advancing Inclusive Mentoring Program**

- Completed training program including 12+ hours of content and discussion about positive and inclusive mentoring practices.

Awards & Fellowships

Sept. 2022 - present **Brinson Prize Fellowship**, Carnegie Observatories / The Brinson Foundation

2022 **Doxsey Travel Prize**, American Astronomical Society

2020 **Student Observing Support**, National Radio Astronomy Observatories (NRAO)

Awards & Fellowships (continued)

2017 — **Summer Research Fellowship**, University of California, San Diego

2016 — **Physics Excellence Award**, University of California, San Diego

2014 — **President's Honor List**, California Polytechnic State University, San Luis Obispo

2011 — **Cal Poly Engineering Scholarship**, California Polytechnic State University, San Luis Obispo

Observing Time

2023-2026 — **Magellan 6.5m Telescopes** (Principal Investigator)
- 15 nights awarded over 6 semesters utilizing new optical IFS (IFU-M) for nearby galaxy survey

2025 — **Local Volume Mapper Open Tile Call** (Co-Investigator)
- 7 hours for study of a nearby galaxy with strong galactic winds

2018-2021 — **W. M. Keck Observatory** (Co-Investigator)
- Led 4.5 nights over 4 semesters with OSIRIS and KCWI resulting in 3 first author publications
- >15 additional nights awarded for programs as a collaborator (OSIRIS, MOSFIRE, KCWI)

2019 (Cycle 7) — **ALMA** (Principal Investigator)
- 14 hour program awarded with $25k funding (observations not completed due to COVID-19 shutdown)

Service & Outreach

Department Service

August 2023-2025 — **Carnegie Observatories Lunch Seminar Committee**
- Member of committee responsible for organizing weekly lunch seminars by: soliciting speaker nominations, selecting speakers, scheduling talks, and coordinating hosts

Sept. 2023 - June 2025 — **Carnegie Observatories Postdoc Representative**
- Liaison between postdocs and Carnegie leadership responsible for organizing regular meetings with both groups, coordinating other postdoc roles, and organizing postdoc events
- Worked with staff to develop a new mentoring program for the postdocs to increase interaction with more staff members

Astronomy Community Service

NSF Review Panels
- Invited participant on multiple NSF Merit Review Panels
- Responsible for grading proposals, writing reviews, and making funding recommendations to program director

Science Organizing Committee Chair – 2025 LVM Science Workshop
- Organized scientific program and funding awards for travel

Time Allocation Committee – LVM Open Tile Call 2025

Manuscript Referee – AAS Journals

Service & Outreach (continued)

Outreach

2022-present — **Carnegie Observatories Public Events**
- Led various tours, talks, and demonstrations for school visits and Observatories Open House

April 2024 — **Perot Museum / Carnegie Science Eclipse partnership**
- Traveled to Dallas for a week of eclipse related education leading up to the 2024 total solar eclipse
- Talked to hundreds of students at local schools and community members at museum events; gave a public talk on eclipse science; emceed an eclipse viewing event to guide safe viewing and provide scientific background in real time

2019-2022 — **Cosmic Tours Portable Planetarium co-coordinator**
- Co-organized the UCSD OIR Lab's portable planetarium program, Cosmic Tours, bringing a ∼25ft diameter inflatable planetarium to K-12 schools and events in San Diego County to provide an exciting astronomy education experiencing
- Coordinated event requests and scheduling presenters, trained new volunteers, and organized team meetings in addition to giving planetarium shows
- Led the creation of a virtual planetarium program to provide an engaging learning experience for students during COVID-19

Presentations

Talks

July 8, 2026 — **SPIE: Astronomical Telescopes + Instrumentation**
Update on the Magellan InfraRed Multi-Object Spectrograph (MIRMOS)

December 9, 2025 — **The Baryon Cycle From Reionization to Cosmic Noon**
Invited Talk: The Magellan InfraRed Multi-Object Spectrograph (MIRMOS)

September 29, 2025 — **Carnegie Science Day**
Studying Stellar Feedback with LVM

June 3, 2025 — **SDSS-V Collaboration Meeting**
Studying Stellar Feedback with LVM: SMC HII Region N66

May 6, 2025 — **Magellan Science Meeting**
The Magellan InfraRed Multi-Object Spectrograph (MIRMOS)

November 18, 2024 — **Carnegie Science Day**
Magellan InfraRed Multi-Object Spectrograph (MIRMOS)

October 23, 2024 — **New Young Researchers in Instrumentation for Astronomy**
A Novel Freeform Slicer IFU for the Magellan InfraRed Multi-Object Spectrograph (MIRMOS)

August 9, 2024 — **CASSI Research Talk Series**
Designing Instruments for the World's Largest Telescopes

June 5, 2024 — **SDSS-V Collaboration Meeting**
Invited Talk: The Local Volume Mapper (LVM) 101

April 7, 2024 — **Public Talk @ Marriot Dallas/Addison Quorum**
The Science of Eclipses

October 26, 2023 — **Surveying the Milky Way: The Universe in Our Own Backyard**
The SDSS-V Local Volume Mapper

July 17, 2023 — **New Views on Feedback and the Baryon Cycle**
Studying Kinematics and Feedback in Local Group H II Regions

Presentations (continued)

June 29, 2023 — **CASSI Research Talk Series**
Combining Observations & Instrumentation

June 15, 2022 — **The 240th American Astronomical Society Meeting**
The Properties and Evolution of Star Forming Regions Over Cosmic Time

February 19, 2019 — **Lorentz Center Meeting: Formation of Stars and Massive Clusters in Dwarf Galaxies over Cosmic Time**
The Size-Luminosity Scaling Relations of Local and Distant Star Forming Regions

January 31, 2019 — **Extremely Big Eyes on the Early Universe, UCLA**
Shedding Light on the Size-Luminosity Scaling Relations of Local and Distant Star Forming Regions

January 7, 2019 — **The 233rd American Astronomical Society Meeting**
IC-10 3D: An IFS Survey of H II *Regions in Local Starburst Galaxy IC-10*

Posters

July 5-10, 2026 — **SPIE: Astronomical Telescopes + Instrumentation**
Update on the slicer IFU for the Magellan infrared multiobject spectrograph (MIRMOS)

July 16-21, 2024 — **SPIE: Astronomical Telescopes + Instrumentation**
A novel freeform slicer IFU for the Magellan InfraRed Multi-Object Spectrograph (MIRMOS)

July 17-22, 2022 — **SPIE: Astronomical Telescopes + Instrumentation**
Liger at Keck Observatory: Imager Detector and IFS Pick-off Mirror Assembly

September 9-10, 2021 — **Keck Science Meeting**
Kinematics and Feedback of H II *Regions in the Dwarf Starburst IC 10*

December 14-18, 2020 — **SPIE: Astronomical Telescopes + Instrumentation**
Liger for Next Generation Keck AO: Filter Wheel and Pupil Design

September 24-25, 2020 — **Keck Science Meeting**
IC-10 3D: Properties of H II *regions in Nearby Starburst Galaxy IC-10*

September 20-21, 2019 — **Keck Science Meeting**
IC-10 3D: An IFS Survey of H II *Regions in Local Starburst Galaxy IC-10 with KCWI*

June 11, 2018 — **SPIE: Astronomical Telescopes + Instrumentation**
Panoramic optical and near-infrared SETI instrument: prototype design and testing

Publications

- 9 first author publications (3 journal articles, 6 SPIE proceedings)
- 3 second author publications (2 journal articles, 1 SPIE proceedings)
- For full list of papers see ADS

Journal Articles

1. **Cosens, M.**, Wright, S. A., Sandstrom, K., Armus, L., Murray, N., Runco, J. N., Sabhlok, S., & Wiley, J. (2024). Oxygen Abundance Throughout the Dwarf Starburst Galaxy IC 10. *AJ*, *168*(6), arXiv 2409.09020, 250. https://doi.org/10.3847/1538-3881/ad7f3c

2. Ocker, S. K., & **Cosens, M.** (2024). Probing the Low-velocity Regime of Nonradiative Shocks with Neutron Star Bow Shocks. *ApJL*, *975*(2), arXiv 2409.13525, L31. https://doi.org/10.3847/2041-8213/ad87cf

3. **Cosens, M.**, Wright, S. A., Murray, N., Armus, L., Sandstrom, K., Do, T., Larson, K., Martinez, G., Sabhlok, S., Vayner, A., & Wiley, J. (2022). Kinematics and Feedback in H II Regions in the Dwarf Starburst Galaxy IC 10. *ApJ*, *929*(1), arXiv 2202.04098, 74. https://doi.org/10.3847/1538-4357/ac52f3

4 **Cosens**, **M.**, Wright, S. A., Mieda, E., Murray, N., Armus, L., Do, T., Larkin, J. E., Larson, K., Martinez, G., Walth, G., & Vayner, A. (2018). Size-Luminosity Scaling Relations of Local and Distant Star-forming Regions. *ApJ*, *869*(1), arXiv 1810.10494, 11. https://doi.org/10.3847/1538-4357/aaeb8f

5 Runco, J. N., **Cosens**, **M.**, Bennert, V. N., Scott, B., Komossa, S., Malkan, M. A., Lazarova, M. S., Auger, M. W., Treu, T., & Park, D. (2016). Broad Hβ Emission-line Variability in a Sample of 102 Local Active Galaxies. *ApJ*, *821*(1), arXiv 1603.00035, 33. https://doi.org/10.3847/0004-637X/821/1/33

SPIE Proceedings

1 **Cosens**, **M.**, Konidaris, N. P., II, Rudie, G. C., Newman, A. B., Aslan, L., Barkhouser, R., Birk, C., Brady, J., Gumy, M., Hare, T., Hope, S. C., Hull, C., Kaismoune, K., Kelson, D. D., Killion, G., Lanz, A., McCloskey, J., Ramirez, S. V., Rupf, C., … Williams, J. E. (2026). Update on the Magellan InfraRed Multi-Object Spectrograph (MIRMOS), In *Ground-based and airborne instrumentation for astronomy xi*, SPIE. https://doi.org/10.48550/arXiv.2607.14052

2 **Cosens**, **M.**, Schurter, P., Konidaris, N. P., Rudie, G. C., Newman, A. B., Aslan, L., Barkhouser, R., Birk, C., Brady, J., Hare, T., Hope, S. C., Hull, C., Kaismoune, K., Kelson, D. D., Killion, G., Lanz, A., McCloskey, J., Ramirez, S. V., Schoenell, W., … Williams, J. E. (2026). Update on the slicer IFU for the Magellan InfraRed Multi-Object Spectrograph (MIRMOS), In *Ground-based and airborne instrumentation for astronomy xi*, SPIE. https://doi.org/10.48550/arXiv.2607.14050

3 Ramrez, S. V., **Cosens**, **M.**, Konidaris, N. P., Garzon, F., McLeod, B., & Ramsay, S. (2026). Community Lessons from builders of infrared multi-object spectrographs, In *Modeling, systems engineering, and project management for astronomy xii*.

4 **Cosens**, **M.**, Konidaris, N. P., Rudie, G. C., Newman, A. B., Killion, G., Aslan, L., Barkhouser, R., Bianco, A., Birk, C., Brady, J., Frangiamore, M., Hare, T., Hope, S. C., Kelson, D. D., Lanz, A., Ramrez, S., Smee, S. A., Vanella, A., & Williams, J. E. (2024). A novel freeform slicer IFU for the Magellan Infrared Multi-Object Spectrograph (MIRMOS) (J. J. Bryant, K. Motohara, & J. R. D. Vernet, Eds.). In J. J. Bryant, K. Motohara, & J. R. D. Vernet (Eds.), *Ground-based and airborne instrumentation for astronomy x*, SPIE. https://doi.org/10.1117/12.3019277

5 **Cosens**, **M.**, Wright, S. A., Brown, A., Fitzgerald, M., Johnson, C., Jones, T., Kassis, M., Kress, E., Kupke, R., Larkin, J. E., Magnone, K., McGurk, R., Rundquist, N.-E., Sohn, J. M., Wang, E., Wiley, J., & Yeh, S. (2022). Liger at Keck Observatory: image detector and IFS pick-off mirror assembly (C. J. Evans, J. J. Bryant, & K. Motohara, Eds.). In C. J. Evans, J. J. Bryant, & K. Motohara (Eds.), *Ground-based and airborne instrumentation for astronomy ix*, SPIE. https://doi.org/10.1117/12.2630219

6 **Cosens**, **M.**, Wright, S. A., Arriaga, P., Brown, A., Fitzgerald, M., Jones, T., Kassis, M., Kress, E., Kupke, R., Larkin, J. E., Lyke, J., Wang, E., Wiley, J., & Yeh, S. (2020). Liger for next-generation Keck AO: filter wheel and pupil design (C. J. Evans, J. J. Bryant, & K. Motohara, Eds.). In C. J. Evans, J. J. Bryant, & K. Motohara (Eds.), *Ground-based and airborne instrumentation for astronomy viii*, SPIE. https://doi.org/10.1117/12.2561837

7 **Cosens**, **M.**, Maire, J., Wright, S. A., Antonio, F., Aronson, M., Chaim-Weismann, S. A., Drake, F. D., Horowitz, P., Howard, A. W., Raffanti, R., Siemion, A. P. V., Stone, R. P. S., Treffers, R. R., Uttamchandani, A., & Werthimer, D. (2018). Panoramic optical and near-infrared SETI instrument: prototype design and testing (C. J. Evans, L. Simard, & H. Takami, Eds.). In C. J. Evans, L. Simard, & H. Takami (Eds.), *Ground-based and airborne instrumentation for astronomy vii*, SPIE. https://doi.org/10.1117/12.2314252